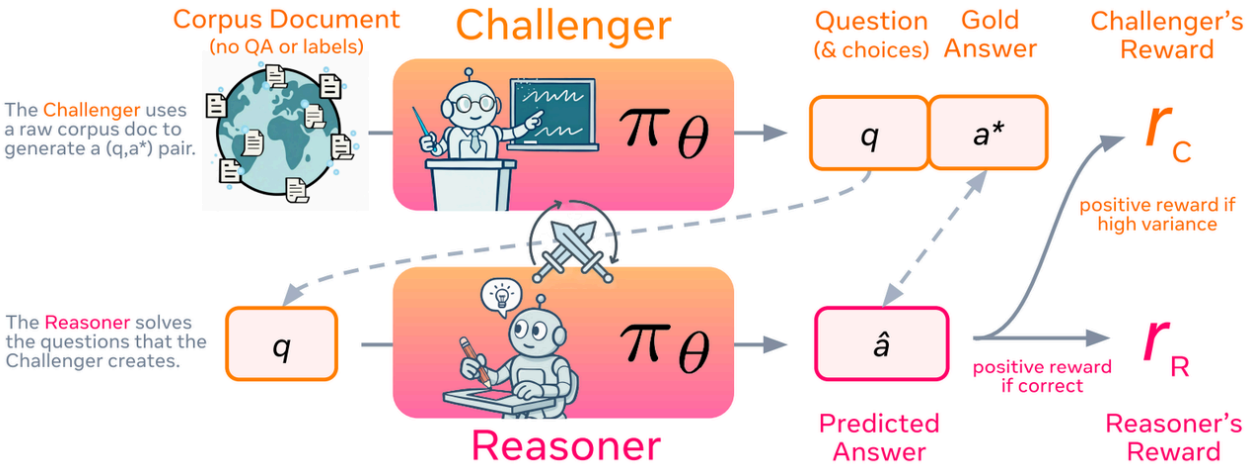
SPICE 🌶️: Self-Play In Corpus Environments Improves Reasoning

- Self-improving systems require environmental interaction for continuous adaptation. **SPICE** is a reinforcement-learning-based adversarial framework to realize self-improvement, where a single model will play two roles: *the Challenger* and *the Reasoner*.



  - Challenger C: When the model is in the Challenger position, it will propose questions for the Reasoner based on web documents.
  - Reasoner R: After being Challenger, the model will be reasoners to answer the questions proposed by the challenger.
- Methodology:
  - Challenger:
    - Challenger's job is to maximize the difficulty of the question while ensuring it's still solvable.
    - We sample segments from a large corpus D, denoted as $d \in D$, from where the questions come from.
    - Will let the challenger propose some questions from the document $d$along with the ground truth:

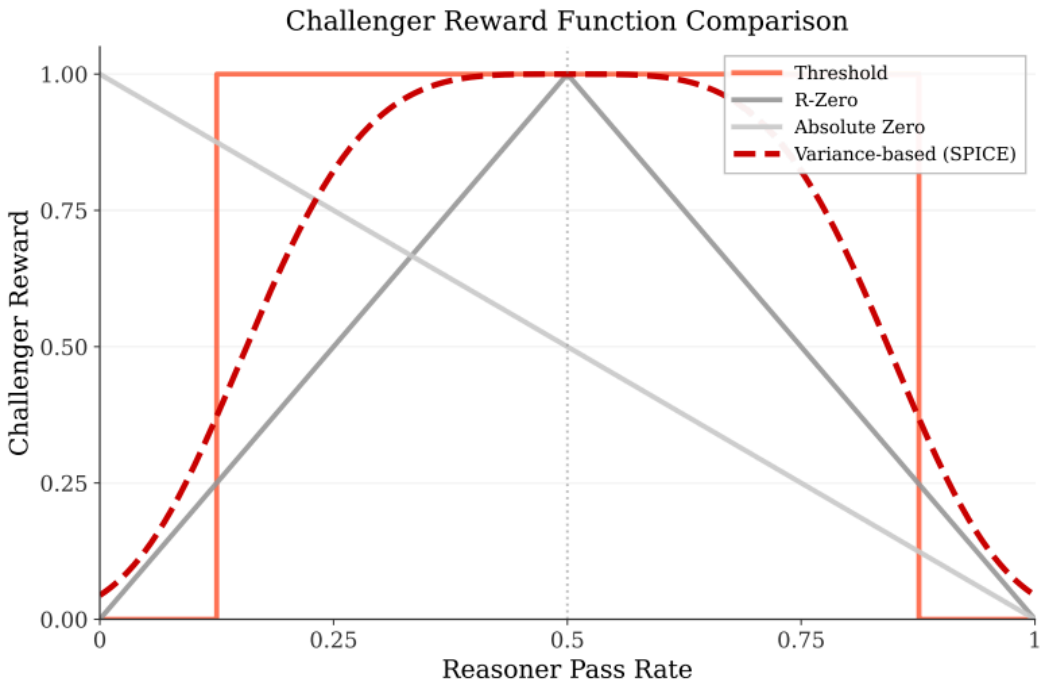
      $(q, a^*) = \pi_\theta(\cdot|d, \text{role} = \text{C})$
    - The questions can be either MCQ or FRQ, and a question is *valid* iff it is in the correct format and parsable.
    - For each valid question q, we let the reasoner generate Kresponse, denoted as $\{\hat{a}_i\}_{i=0}^K$. Based on the correctness of answers, we generate a reward for the challenger:

      $$r_C(q, a^*) = \begin{cases} \exp\left(-\frac{(\text{Var}(\{l_1,\ldots,l_K\})-0.25)^2}{2\cdot 0.01}\right) & \text{if } q \text{ is valid} \\ \rho & \text{otherwise (penalty)} \end{cases},$$

      where $l_i = \mathbb{I}[\hat{a}_i = a_i^*]$.
    - Note that our goal is to let the question be **DIFFICULT** yet **SOLVABLE.** For example, if a set of answers to a question contain 50% correct answers, meaning that the reasoner believes that the question is hard but still is able to handle. If every answer is wrong or right, then

the question could be either unsolvable or too easy. In that case, this is NOT a good question.

- With this idea in mind, the variance-Gaussian reward function is used. A high variance means the reasoner is not sure, i.e., 50% wrong-50% correct, whereas a low variance indicates "unsolvable or too easy" (all correct or all incorrect).



- The figure above is from the ablation experiment. Four functions are tested.
  - *Absolute Zero* has the worst performance, as it actually encourages the challenger to give out easy questions.
  - *Threshold* has the 2nd worst performance. Though it encourages "difficult yet solvale", it has two fatal problems: it is not continuous, and in some ranges (two ends), there can be no training signals.
  - *R-Zero* performs better, but it's not differentiable.
  - *SPICE* is the best. It fulfills "difficult yet solvale", and it is differentiable and smooth.

Challenger Reward	Math	General	Overall
Absolute Zero	48.2	30.8	40.7
Threshold	48.6	31.6	41.4
R-Zero	50.0	33.9	43.6
Variance (SPICE)	**50.6**	**35.0**	**44.9**

- Reasoner
  - Given a valid question q, generate answer $\hat{a} \sim (\cdot|q, \text{role} = \text{R})$.
  - Reward also uses $l_i = \mathbb{I}[\hat{a}_i = a_i^*]$.
- Training:
  - We use `DrGRPO`.

- Loss function:

  $$J(\theta) = \mathbb{E}_{d\sim\mathcal{D}}\left[\mathbb{E}_{(q,a^*)\sim\pi_\theta(\cdot|d,\text{role}=C)}[r_C(q,a^*)] + \mathbb{E}_{\hat{a}\sim\pi_\theta(\cdot|q,\text{role}=R)}[r_R(\hat{a},a^*)]\right]$$

- Advantages:

  $\hat{A}_C^i = r_C^i - \text{mean}(\{r_C^j\}_j)$
  $\hat{A}_R^i = r_R^i - \text{mean}(\{r_R^j\}_j)$, similar to that of `GRPO`, but `DrGRPO` has no standard deviation normalization.

- Pseudocode

Algorithm 1 SPICE 🌶️: Self-Play In Corpus Environments

Require: Pretrained LLM π_θ; corpus $\mathcal{D}$; batch size B; group size G; iterations T; penalty ρ

for $t \leftarrow 1$ to T **do**
  Challenger Role: Generate challenging problems with document access
  for $b \leftarrow 1$ to B **do**
    Sample document $d \sim \mathcal{D}$
    Generate multiple attempts: $\{(q_i, a_i^*)\}_{i=1}^N \leftarrow \pi_\theta(d, \text{role} = C)$ ▷ MCQ or free-form
    $\mathcal{T}_C \leftarrow$ Subsample G trajectories preserving valid:invalid ratio
    if $q_i \in \mathcal{T}_C$ is valid **then**
      $\{\hat{a}_k\}_{k=1}^G \leftarrow \pi_\theta(q_i, \text{role} = R)$ ▷ No document access
      $r_C(q_i, a_i^*) \leftarrow$ Compute reward using Eq. (1) ▷ Gaussian variance reward
    else
      $r_C(q_i, a_i^*) \leftarrow \rho$ ▷ Invalid task penalty
    end if
  end for
  Reasoner Role: Solve generated problems without document access
  Select a random valid task (q, a^*) from Challenger phase
  $\mathcal{T}_R \leftarrow \{\hat{a}_i\}_{i=1}^G \sim \pi_\theta(q, \text{role} = R)$ ▷ G responses for training
  for $i \leftarrow 1$ to G **do**
    $r_R(\hat{a}_i, a_i^*) \leftarrow \mathbb{1}[\hat{a}_i = a^*]$ ▷ Binary correctness reward
  end for
  Update Phase: Optimize π_θ with role-specific advantages
  for trajectory i in $\mathcal{T}_C$ **do**
    $\hat{A}_C^i \leftarrow r_C^i - \text{mean}(\{r_C^j : j \in \mathcal{T}_C\})$ ▷ DrGRPO
  end for
  for trajectory i in $\mathcal{T}_R$ **do**
    $\hat{A}_R^i \leftarrow r_R^i - \text{mean}(\{r_R^j : j \in \mathcal{T}_R\})$ ▷ DrGRPO
  end for
  Update π_θ via policy gradient with advantages $\{\hat{A}_C^i\}$ and $\{\hat{A}_R^i\}$
end for
return Trained model π_θ

- Experiment
  - **Superior Performance**: SPICE consistently outperformed all baselines (including R-Zero and Strong Challenger) across four model families. It achieved significant gains in both mathematical (+8.9%) and general reasoning (+9.8%) benchmarks.
  - **Adversarial Dynamics**: The experiments validated "co-evolution." As training progressed, the Challenger generated increasingly difficult tasks (lowering a fixed Reasoner's pass rate from 55% to 35%), while the Reasoner's ability to solve fixed tasks improved (raising pass rate from 55% to 85%).
  - **Critical Components**: Ablation studies proved that **Variance-based rewards** provided the best training signal compared to R-Zero or Threshold rewards. Additionally, combining **MCQ and Free-form** task formats yielded better results than using either format alone.
  - **Corpus Grounding**: Removing the corpus (pure self-play) caused performance to drop significantly, confirming that external documents are essential for sustained improvement.
- Work Cited
  - [Paper]

- Chat History